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Can the Simple View of Reading Inform the Study of Reading Comprehension in Young Spanish Heritage Language Learners?

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ABSTRACT

This study examines the development of reading comprehension (RC) in Spanish heritage language (HL) learners in the Midwest United States. While extensive research exists on RC in monolingual English-speaking populations, applying reading science models such as the Simple View of Reading (SVR) in HL or bilingual contexts remains under-researched. Additionally, orthographies labeled as phonologically transparent—where grapheme-to-phoneme correspondences are relatively more consistent—such as Spanish, have not been thoroughly investigated within these frameworks. This study applies the SVR framework to a unique population of 74 Spanish-English bilingual children, centering the Spanish HL rather than the majority language at two reading stages: early readers and pre-readers. Using Structural Equation Modeling, we assess (1) concurrent roles of decoding and linguistic comprehension as SVR's cognitive bases for RC development in early readers and (2) exploratory relationships between decoding and linguistic comprehension predictors in pre-readers. Results for Spanish HL readers indicate that linguistic comprehension, but not decoding, predicts significant unique variance in Spanish RC; however, the two components share significant variance and, combined, explain most of the variance in RC. Results for Spanish HL pre-readers show a high correlation between linguistic comprehension and decoding predictors, indicating the interwoven nature of the two SVR components at earlier reading stages. The findings advance the discussion on the relevance of the SVR in phonologically transparent orthographies and across different stages of reading development, bridging minority and majority language research and informing educational strategies for Spanish HL learners.

As the ultimate developmental outcome of reading, reading comprehension (RC) is an essential skill for successful learning and cultural integration. This study examines young Spanish heritage language (HL) learners' cognitive skills associated with RC in Spanish at two stages of reading development: the pre-reading (emergent literacy) and the early reading stage.

Given that RC engages over 30 cognitive and metacognitive processes (Randi et al. 2005), the strength of the Simple View of Reading (SVR; Gough and Tunmer 1986; Hoover and Gough 1990) lies in its parsimony, testability (Shanahan 2023), and capacity for informing instruction by identifying children's strengths and weaknesses based on their word reading and linguistic comprehension skills (Kim 2023). As a skill-based model, the SVR distinguishes itself from frameworks such as the Construction–Integration model (Kintsch and Van Dijk 1978) and the Reading Systems Framework (Perfetti and Stafura 2014), which emphasize global cognitive processes (rather than skills) involved in constructing mental representations during reading.

The SVR is well-supported in monolingual English-speaking populations (e.g., García and Cain 2014; Hjetland et al. 2020; Ripoll Salceda et al. 2014). Although quantitative SVR research in phonologically transparent orthographies (orthographies with relatively consistent phoneme-grapheme mapping, Van den Bosch et al. 1994) remains limited, existing findings align with the robust evidence from English. Across both types of orthographies, decoding and linguistic comprehension together explain approximately 60% of the variance in RC (Joshi et al. 2012; Lee et al. 2022; Ripoll Salceda et al. 2014). Variations between the so-named phonologically transparent and opaque orthographies highlight differences in the contributions of decoding and comprehension to RC and how these factors evolve during children's literacy development. The simplified nature of SVR thus provides a principled foundation for exploring these important questions.

Alternative models have been proposed to address SVR limitations by adding specific underlying mechanisms for word reading and linguistic comprehension and incorporating higher order cognitive and sociocultural factors to influence RC (see reviews in Duke and Cartwright 2021; Kim 2023). By adding levels of complexity, these frameworks offer insights into the intricacies of reading development (Kim 2020; Aaron et al. 2008); however, we argue that building onto SVR will be most productive once a clearer performance benchmark of the baseline SVR model has been established—particularly in understudied groups such as HL and bilingual learners. This study contributes novel evidence from Spanish HL learners with an eye to deepening understanding of the baseline SVR model.

A HL learner in the United States is typically an early bilingual who has been exposed to the HL since birth and to English simultaneously or later in development, often without complete acquisition of either language (Benmamoun et al. 2013). For Spanish HL learners, Spanish is often embedded in daily life, serving as the primary language at home and a key medium for family and social interactions (Valdés 2005). RC in Spanish extends beyond academic settings, facilitating active engagement in cultural and social activities such as religious services and classes, text messaging within the HL community, and consuming Spanish-language media. Developing literacy in Spanish not only strengthens these interactions but also reinforces a sense of identity and belonging (Arredondo et al. 2016; Montero-Sieburth and Barth 2001). Moreover, cross-linguistic research highlights that HL skills can support second language (L2) English literacy development through cross-linguistic transfer (Cummins 2005; Melby-Lervåg and Lervåg 2011; Prevoo et al. 2016; Proctor et al. 2017; Wagley et al. 2022).

With Hispanic students making up about 29% of the US school population and often lagging behind national standards in reading (National Center for Education Statistics 2025a, 2025b), it is essential to conduct research that sheds light on their unique linguistic needs. Given the significant influence of the SVR on educational policy for English monolinguals (Hoover and Tunmer 2018), assessing its applicability to HL contexts ensures evidence-based relevance and accuracy. By moving away from a monolingual English standard, this investigation focuses on young HL children's challenging task of developing Spanish RC within an English-dominant environment.

1 | The SVR and Its Components

Within the SVR model, RC is defined as the ability to derive and construct implicit and explicit meaning from printed text. RC is typically assessed through open-ended tasks, such as short-answer questions and retelling, or closed-ended tasks, including cloze (e.g., fill-in-the-blank) and multiple-choice formats (Baker et al. 2022; Florit and Cain 2011; Ripoll Salceda et al. 2014). The SVR formulaically expresses the individual and combined contribution of decoding and linguistic comprehension to RC as $R = \text{Decoding} \times \text{linguistic comprehension}$ (Hoover and Gough 1990). If either of the factors is assessed as zero mastery, the RC product will also result in zero.

Decoding is constrained by print-based processing and is particular to reading. Specifically, the process is defined as efficient word recognition involving the accurate and rapid recognition of printed words to access word meanings stored in the internal mental lexicon (Hoover and Tunmer 2018). In the broader reading literature, decoding is often described as the process of word recognition without necessarily including the retrieval of meaning. The SVR assumes RC requires a decoding system, such as phonological coding (i.e., letter-sound correspondence), that facilitates efficient retrieval of word meanings. Decoding is typically measured as fluency and accuracy of word and nonword recognition (Ripoll Salceda et al. 2014).

In contrast to decoding, linguistic comprehension involves deriving meaning from spoken language (Hoover and Tunmer 2018). This process demands a wider array of language skills due to the complex nature of spoken communication (Lervåg et al. 2018). The SVR originally equated linguistic comprehension with listening comprehension (Gough and Tunmer 1986; Hoover and Gough 1990), without detailing its underlying skills. However, the growing use of latent variable modeling has enabled a more nuanced examination of the multiple skills involved (Cadime et al. 2017; Language and Reading Research Consortium and Chiu 2018; Lee et al. 2022; Lervåg et al. 2018), moving beyond the traditional reliance on listening comprehension.

This study focuses on vocabulary (i.e., lexicon) and morphosyntax (i.e., grammar) as key indicators of linguistic comprehension. We draw upon abundant research that pinpoints grammar and vocabulary as fundamental oral language skills (Alonzo et al. 2016; Kim 2015, 2016; Lepola et al. 2012). Vocabulary is critical for RC as it provides access to word meanings even before learning to read (Protopapas et al. 2013). Extensive data across interventions (e.g., Elleman et al. 2009; Fricke et al. 2013; Stahl and Fairbanks 1986), individual differences (e.g., Colenbrander et al. 2016; Joshi 2005), and longitudinal studies (e.g., Quinn et al. 2015; Storch and Whitehurst 2002; Verhoeven and Van Leeuwe 2008) consistently underscore the pivotal role of vocabulary in RC. Morphosyntax also supports RC by facilitating sentence processing, integration of information, and inference of unfamiliar words (Cain and Oakhill 2006; Jeon and Yamashita 2014). Zheng et al.'s (2023) meta-analysis demonstrates a robust correlation between morphosyntax and RC across first (L1) and L2 language studies from primary school to university. Research further highlights the prominent role of morphosyntax and vocabulary in RC among bilingual elementary school children learning an L2 with a phonologically transparent orthography (e.g., Bellocchi et al. 2017; Junyent and Fernández-Flecha 2022), with vocabulary emerging as a stronger predictor of RC in L2 than in L1 (Droop and Verhoeven 2003; Lervåg and Aukrust 2010).

Despite growing recognition of morphosyntax as a key component of linguistic comprehension in SVR studies, grammatical features remain less explored than vocabulary. For instance, only 25% of the predominantly English-language studies reviewed in meta-analyses by Hjetland et al. (2020) and Ripoll Salceda et al. (2014) included measures of morphosyntax, compared to 68% for vocabulary. Research on Spanish morphosyntax within the SVR framework is also meager, with Cubilla Bonnetier and Sánchez-Vincitore (2025) and Wagley et al. (2022) as the only studies identified.

To bridge this gap, our study is the first to explore how a latent factor combining vocabulary and morphosyntax contributes to RC within the SVR framework, focusing on children's HL Spanish at two stages of reading development. We prioritize foundational oral language skills that are particularly relevant to bilingual and HL learners. An advantage of this strategy, as opposed to using a single measure such as listening comprehension, is that it offers a more precise understanding of the skills to target in instruction for developing Spanish RC.

1.1 | The SVR Components in Pre-Readers

Despite SVR being initially conceived to take into account decoding and linguistic comprehension concurrently, many researchers have examined emergent decoding and linguistic comprehension skills as predictors of later elementary RC in predominantly monolingual English samples (e.g., Hjetland et al. 2020). These studies often consider decoding precursors, including letter naming and phonemic/phonological awareness. Meanwhile, linguistic comprehension predictors usually cover vocabulary, grammar, and listening comprehension (Hjetland et al. 2017).

Research using exploratory and confirmatory factor analyses consistently shows that code-related and linguistic comprehension skills are distinct factors in preschoolers learning to read in both English (Catts et al. 2015; Hjetland et al. 2020; Kendeou, Savage, et al. 2009; Language and Reading Research Consortium and Chiu 2018) and Greek (Kendeou et al. 2013; Protopapas et al. 2012). In English, the two components are strongly correlated with values ranging from 0.70 to 0.84 (Hjetland et al. 2020; Language and Reading Research Consortium and Chiu 2018; Storch and Whitehurst 2002), whereas in Greek, the correlation is weaker, ranging from 0.17 to 0.37 (Kendeou et al. 2013; Protopapas et al. 2012). Overall, research has not examined the separability of SVR components in phonologically transparent orthographies in pre-readers. Investigating this question in Spanish HL pre-readers extends prior work by evaluating the framework's validity at an earlier stage of development. This study asks how decoding and linguistic comprehension relate in pre-readers versus early readers, a comparison that remains largely unexplored, particularly among Spanish monolingual and HL learners.

2 | SVR in Phonologically Transparent Orthographies

In alphabetic languages, the so-named phonologically transparent or shallow orthographies have a generally consistent or high predictability mapping between phonemes and graphemes (e.g., Italian, Indonesian, Greek, and Finnish), as opposed to opaque or deep orthographies in which the same letter or group of letters (graphemes) can have multiple pronunciations, as it occurs in English, French, or Danish (Van den Bosch et al. 1994).

There is substantial support for the developmental trend in English-speaking populations where decoding initially has a significant impact on RC, but linguistic comprehension becomes more influential as children progress in their reading development (Florit and Cain 2011; García and Cain 2014; Hjetland et al. 2020). Conversely, research on phonologically transparent orthographies consistently shows that linguistic comprehension, rather than decoding, has a stronger influence on RC. This pattern is observed in early reading and persists throughout developmental stages for children aged 6–11 in Italian ($N = 149$, Florit et al. 2022; $N = 1895$, Tobia and Bonifacci 2015), Portuguese ($N = 325$, Cadime et al. 2017), and Finnish ($N = 1815$, Torppa et al. 2016). The stronger role of linguistic comprehension is often attributed to lower decoding complexity in phonologically transparent orthographies, whereas the complexity of opaque orthographies arguably requires a heightened reliance on decoding for RC and more time to master overall (Tobia and Bonifacci 2015).

Interestingly, recent research on Spanish, traditionally classified as a phonologically transparent orthography, reveals a different pattern: Spanish monolinguals aged 6–10 show a pattern similar to English monolinguals, with decoding exhibiting a stronger relationship with RC than linguistic comprehension. This finding is supported by Baker et al.'s (2022) meta-analysis including children aged 5–8, along with individual studies from Mexico ($N = 53$, ages 6–7; Camarillo Salazar et al. 2021), the Dominican Republic ($N = 4750$, ages 8–10; Sánchez-Vincitore et al. 2022), Chile ($N = 350$, ages 6–7; Kim and Pallante 2012), and Panama ($N = 113$, ages 7–8; Cubilla Bonnetier and Sánchez-Vincitore 2025).

However, these Spanish monolingual studies focus on decoding fluency, using measures such as nonword reading fluency and reading speed, rather than considering both accuracy and fluency, as seen in the phonologically transparent orthographies studies reviewed above. In phonologically transparent orthographies, research has shown that decoding fluency initially plays a dominant role, surpassing linguistic

TABLE 1 | Summary of the skills assessed for each SVR component and group of readers.

	Decoding	Linguistic comprehension	RC
SVR definition (Hoover and Tunmer 2018)	The ability to accurately and rapidly recognize printed words to access word meanings stored in the internal mental lexicon	The ability to derive meaning from spoken language	The ability to derive and construct implicit and explicit meaning from printed text
Pre-readers skills	Precursors: letter recognition, phonemic awareness and letter-sound correspondence	Receptive vocabulary Morphosyntax	
Early readers skills	Accuracy and fluency (i.e., accuracy rate and words correct per minute, respectively)	Receptive vocabulary Morphosyntax	Retelling task

comprehension. However, as children advance through schooling, by third grade and beyond, linguistic comprehension takes precedence over decoding accuracy and fluency (Baker et al. 2022; Florit and Cain 2011; Torppa et al. 2016). In our study, we assess decoding by incorporating both a reading fluency and a reading accuracy measure, aligning with the SVR's definition of decoding, which encompasses both dimensions, efficiency and accuracy.

Beyond the monolingual literature, the limited research on Spanish RC in Spanish-English bilinguals shows mixed results regarding the roles of decoding and linguistic comprehension (Gottardo et al. 2014; Joshi et al. 2012; Wagley et al. 2022). While these studies are relevant as they provide insight into Spanish RC in HL learners, they are limited in scope. Most rely on regression analyses and do not include fluency measures, making comparisons with earlier Spanish monolingual studies difficult. Only Wagley et al. (2022) and Goodrich and Namkung (2019) have employed path analysis, with Wagley et al. (2022) being the only study to use latent modeling to explore SVR components. Our study addresses these limitations by investigating how a decoding latent factor, capturing both accuracy and fluency, and a linguistic comprehension factor, including vocabulary and morphosyntax, contribute to HL Spanish RC at two distinct stages of reading development.

3 | The Current Study

Adhering to the baseline SVR framework, we examine cognitive skills related to RC in a sample of young Spanish HL learners (ages 4–9) in the Midwest at different stages of reading development (pre-readers and early readers). To ensure clarity in the use of key reference terms, we define early readers' decoding skills as words correct per minute and accuracy. We specify pre-readers' decoding precursors as including letter recognition, letter-sound correspondence, and phonemic awareness. Finally, linguistic comprehension refers to skills in morphosyntax and receptive vocabulary across developmental groups. A summary of these skills is provided in Table 1.

The following hypothesis and research question guide our model:

- **Hypothesis** (early readers): *Spanish decoding and linguistic comprehension significantly and uniquely contribute to Spanish RC in HL early readers.*
- **Exploratory Research Question** (pre-readers): How are decoding precursors and linguistic comprehension skills associated in Spanish HL pre-readers? Given the high correlation reported in the literature during the preschool years, we expect to observe a similarly strong association between decoding and linguistic comprehension in Spanish HL pre-readers.

This study departs from previous SVR research by introducing three novel features: (a) it centers the HL, Spanish, rather than the majority language, in the participant children's reading development, (b) it includes a measure of morphosyntax alongside vocabulary to determine their joint contributions to RC, and (c) it explores the separability of the SVR components and the strength of their association at two stages of reading development in Spanish HL.

4 | Methods

4.1 | Participants

The study participants were 74 Spanish HL speakers, ages 4.69–9.34, who attended a Saturday Spanish HL school in SE Michigan, an area of recent immigrant Latino/*x* families. We categorized children into two groups for the analyses: *early readers* ($N = 42$; $M_{age} = 7.1$, $SD = 0.92$) and *pre-readers* ($N = 32$; $M_{age} = 5.86$, $SD = 0.56$). For differentiating early readers from pre-readers, we established a baseline criterion of the child's ability to read a 16–25-word passage (AA level from the Learning A–Z text leveling system), corresponding with kindergarten (ages 4–6) level and level A in the Fountas and Pinnell leveling system (Heinemann 2024). Children in the early reading group had a Spanish reading ability equivalent to kindergarten (38.1%) and first grade (57.1%).

The Saturday Spanish school class organization mirrors the daily English school's preschool to third-grade age range. Excluding three children diagnosed with specific conditions (i.e., dyslexia, apraxia, and hearing impairment), all participants were typically developing. Participants' families resided primarily in upper-middle-income areas, as averaged zip code data suggested, although these regions encompassed a range of socioeconomic statuses. This study complies with all investigative protocols and has IRB approval from the corresponding institution.

4.2 | Measures

4.2.1 | Linguistic Comprehension: Early Readers and Pre-Readers

4.2.1.1 | Vocabulary. Building on previous work focusing on receptive vocabulary (e.g., Gottardo et al. 2014, 2018; Lonigan et al. 2018; Torppa et al. 2016), we examined children's vocabulary size using the Spanish adaptation of the Peabody Picture Vocabulary Test (PPVT), "Test de Vocabulario de Imágenes Peabody" (TVIP; Dunn et al. 1986), for assessing receptive vocabulary. The authors of the TVIP reported split-half reliability of 0.93 (Dunn et al. 1986), indicating strong internal consistency.

4.2.1.2 | Morphosyntax. We administered the experimental version (field test) of the Bilingual English-Spanish Assessment—Middle Elementary (BESA-ME; Peña et al. 2016) to measure Spanish morphosyntactic skills (namely, irregular past and imperfect past tense forms, relative clauses, subjunctive verb forms, and adjectives). We extend BESA-ME to younger age groups (4.69–7.0) based on the instrument's alignment with the Spanish proficiency levels previously assessed in our study's participants and pilot results from five HL children. The test consists of a cloze and a sentence repetition task. Alpha coefficients for the field test sample can be obtained from Bedore et al. (2018).

4.2.2 | Decoding Precursors: Pre-Readers

4.2.2.1 | Alphabet Recognition, Phonemic Awareness, and Letter-Sound Correspondence. To evaluate pre-readers beginning Spanish reading skills, we implemented the Illinois Snapshots of Early Literacy-K/1-Spanish (ISEL-S; Barr et al. 2004), normed for HL speakers ages 5–7 residing in the US Midwest. We administered letter recognition, phonemic awareness (matching a picture to a word based on initial sounds), and letter-sound associations (matching sounds to letters and digraphs) snapshots from the ISEL-S. While reliability data for the Spanish version were not available, the English version (ISEL-K/1) demonstrated good reliability (Cronbach's alpha > 0.74) across the three components (Barr et al. 2004).

4.2.3 | Decoding: Early Readers

We evaluated early HL readers' accuracy and words correct per minute (WCPM) using the Spanish reading subtest of the Running Record assessment available on the Reading A–Z (Raz-Kids) platform (Lazel Inc. 2024). Participants read a fiction or non-fiction passage on the same general topic, previously assigned by the researcher, with a complexity level (e.g., word count, sentence length, number, and difficulty of unfamiliar words) matching the child's ability. These passages are created specifically to reflect Latino/*x* cultural identity, incorporating culturally relevant settings, characters, and events rather than simple translations of English texts.

Reading errors were counted whenever the child substituted, omitted, inserted, or mispronounced words (excluding dialect-related differences).

4.2.4 | Reading Comprehension: Early Readers

We assessed readers' RC using the Running Record (Lazel Inc. 2024) retelling subtest. Early readers read their assigned Spanish passage and subsequently recounted the details of the engaged text in the HL. Using the Running Record rubric, the retelling task was scored on a 0–3 scale (from incomplete or incorrect to complete). The rubric assessed the use of details, characters, setting, summary, vocabulary, main idea, and precision.

4.3 | Procedures

Data collection took place at the conclusion of two consecutive academic years. The first author and the HL School evaluation team trained research assistants (primarily graduate students who were native Spanish speakers) to administer the tests. Children's reading and retelling tasks were audio-recorded and jointly scored by the first author and at least one Spanish-speaking research assistant to ensure scoring reliability. In early readers, the selection of appropriate passage level was informed by the assessment's benchmarks based on each child's previous performance on the Running Record conducted regularly at the Spanish school. Passage levels from Learning A to Z align with US Common Core State Standards for text complexity (Common Core State Standards Initiative 2021), correlating with established leveling systems like Fountas & Pinnell and Reading Recovery (Lazel Inc. 2024). Each passage level was coded sequentially to facilitate statistical analysis, starting with "1" for Level A, "2" for Level B, and so on, up to "15" for Level N, the highest level assigned. One-third of the participants were assigned to level D, while levels B and F had assignment rates of 14%. The participants' reading of either fiction or non-fiction texts was randomly assigned, with 20 children assigned fiction and 22 assigned non-fiction texts.

4.4 | Data Analysis

To assure data quality, we relied on preliminary analyses that assessed normality and outliers using z-scores ($|z| < 1.96$ for approximate normality, ± 3 for outliers; Field 2009) and identified potential covariates. We used Structural Equation Modeling (SEM) to examine decoding and linguistic comprehension's contribution to RC in early readers. We opted for SEM to evaluate our HL reading model due to its ability to account for measurement error, which can bias results in traditional regression analyses (Cole and Preacher 2014). SEM also enables the use of multiple indicators to represent latent constructs more accurately, improving construct validity. Additionally, SEM's full maximum likelihood estimation is robust to missing data, enhancing the reliability of the findings.

To maintain consistency with our SEM framework for the early readers, we conducted Confirmatory Factor Analysis (CFA) within SEM to examine the relationship between decoding and linguistic comprehension precursors in Spanish HL pre-readers. This approach allowed us to draw preliminary (exploratory) comparisons with the developmentally more advanced (early) readers.

We assessed the model fit with CFI and TLI values of 0.95 or higher and SRMR values below 0.08, which indicate good fit. An excellent fit was indicated by an RMSEA below 0.05, while values below 0.08 suggested a moderate fit, following guidelines from West et al. (2012). We handled missing data using Full Information Maximum Likelihood (FIML), which utilizes all available data to estimate the covariance matrices to be analyzed (Enders and Bandalos 2001). We employed MPlus version 8.18 and SPSS version 29 in data analysis.

We performed MANOVA Box's M to determine whether the covariance among observed variables—set to be included in the SEM model—was stable across different groups of participants. Specifically, we compared the covariance structures of children in the first academic year with those in the second and children who read non-fiction with those who read fiction texts. A non-significant Box's M result indicates that the groups' covariance matrices do not differ substantially. This provides a rationale for not controlling group membership as a covariate within the SEM. Additionally, we examined how children's age and passage level correlated with the variables in the model. We treated the passage's level as an interval scale, assuming each increment in difficulty level represents a consistent increase in complexity.

5 | Results

To recapitulate, this study tests the hypothesis that Spanish decoding and linguistic comprehension significantly and uniquely contribute to Spanish RC in HL early readers, as based on the SVR framework. Additionally, we explore the association of decoding precursors and linguistic comprehension skills in Spanish HL pre-readers. We first report the results for HL early readers, followed by the pre-readers' results.

5.1 | Early Readers

5.1.1 | Preliminary Analysis

Table 2 shows descriptive statistics. We conducted two MANOVAs to investigate the effects of different independent variables—reading passage type and academic year—on a set of dependent variables: morphosyntax, vocabulary, accuracy, and WCPM. The MANOVA Box's M test for equality of covariance matrices yielded nonsignificant results for passage type, $M=10.09$, F (15, 4898) $=0.57$, $p=0.90$; and academic year, $M=9.12$, F (15, 4636) $=0.51$, $p=0.94$. Given these nonsignificant results, we proceeded with the SEM analysis under the assumption that the variance–covariance matrix was equivalent across the levels of passage type and academic year.

Table 3 displays correlations between children's age and passage level and the variables included in the SEM model. Although correlations were in the expected direction (positive correlations; the higher the age/text level, the higher the test score), none were significant. Guided by parsimony, we did not include age, passage level, passage type, and academic year as control variables in our SEM model.

5.1.2 | SEM Analysis

We specified a measurement model defining two latent constructs—"linguistic comprehension" and "decoding"—and a structural model to examine the relationship between the two latent factors and the degree to which each latent factor predicted RC. Linguistic comprehension indicators were receptive vocabulary and morphosyntax scores, while decoding indicators were reading accuracy and WCPM scores. For reasons of model parsimony, residual variance (error terms) was assumed to be equal for indicators of the same construct. The full SEM model with the resulting parameter estimations is shown in Figure 1. Missing data were negligible, with four participants (9.5%) missing data on morphosyntax and one participant (2%) missing data for receptive vocabulary and RC scores.

The measurement model revealed strong factor loadings for the latent factors of linguistic comprehension (0.92 for receptive vocabulary and 0.84 for morphosyntax) and decoding (0.91 for WCPM and 0.67 for accuracy). The correlation between the two latent factors was positive and significant, $r=0.51$, $p<0.001$. Further, the structural model tested the concurrent contributions of decoding and linguistic comprehension on RC. Results showed that linguistic comprehension positively and significantly predicted RC ($\beta=0.63$, $SE=0.12$, $p<0.001$), whereas decoding did not ($\beta=0.20$, $SE=0.14$, $p=0.15$). The model explained 56.6% of the variance in RC. Linguistic comprehension and decoding explained 40% and 4% of the variance in RC, respectively. The overall SEM model evidenced a very good fit to the data (RMSEA = 0.06; SRMR = 0.04; CFI = 0.99; TLI = 0.98). In a further exploratory step, we examined the latent interaction between decoding and linguistic comprehension in predicting RC but found it to be statistically insignificant ($\beta=-0.07$, $SE=0.12$, $p=0.55$).

Employing an alternative analysis, hierarchical regression, yielded similar results, contributing to the robustness of our findings despite potential sample size constraints with SEM.[1]

5.2 | Pre-Readers

5.2.1 | Preliminary Analysis

Table 4 shows descriptive statistics for pre-readers. Prior to the CFA, we conducted correlation analyses between the observed variables that served as indicators for the latent factors (i.e., decoding and linguistic comprehension), as detailed in Table 5.

TABLE 2 | Descriptive statistics for readers ($N = 42$).

	n	%	Min	Max	M (SD)
Child characteristics					
Gender					
Boys	22	52.4			
Girls	20	47.6			
Spanish school grade					
Kindergarten	17	40.5			
First grade	17	40.5			
Second grade	8	19			
Generation					
Year 1	24	57.1			
Year 2	18	42.9			
Age	42		5.82	9.34	7.1 (0.92)
Linguistic comprehension					
Receptive vocabulary (max = 125)	41		25	99	64.17 (19.36)
Morphosyntax (max = 49)	38		0	49	31.21 (13.7)
Decoding					
Accuracy (max = 100)	42		41.46	100	85.76 (13.7)
WCPM	42		3.22	123.67	39.01 (30.31)
Reading comprehension (max = 18)	42		0	18	10.4 (5.32)

Abbreviation: WCPM, words correct per minute.

TABLE 3 | Bivariate correlations in readers.

Variable	1	2	3	4	5	6	7
1. Receptive vocabulary	—						
2. Morphosyntax	0.76***	—					
3. WCPM	0.39*	0.34*	—				
4. Accuracy	0.42**	0.16	0.6***	—			
5. Age	0.26	0.06	0.15	0.05	—		
6. Passage level	0.22	0.26	0.27	0.1	0.35*	—	
7. Reading comprehension	0.67***	0.58***	0.41**	0.45**	0.15	0.19	—

*$p < 0.05$.
**$p < 0.01$.
***$p < 0.001$.

5.2.2 | CFA

We conducted a CFA to explore whether decoding precursors and linguistic comprehension skills constitute two distinguishable components in pre-readers and, if so, how these two components relate to each other. Decoding was indicated by phonemic awareness (PA), letter recognition (LR), and letter-sound correspondence (LS) while listening comprehension was indicated by morphosyntax and vocabulary. Only two variables included missing values: morphosyntax with missing data from three participants (9.3%) and PA for one participant (3.12%). Results from the CFA are presented in Figure 2. The two-factor model demonstrated an excellent fit to the data (RMSEA = 0; SRMR = 0.04; CFI = 1; TLI = 1). All measures loaded significantly on their corresponding factors; factor loadings were strong for morphosyntax, vocabulary, and LR (> 0.79), weak for PA, and moderate for LS (0.37 and 0.66, respectively). The correlation between the two latent factors was positive and significant, $r = 0.83$, $p < 0.001$.

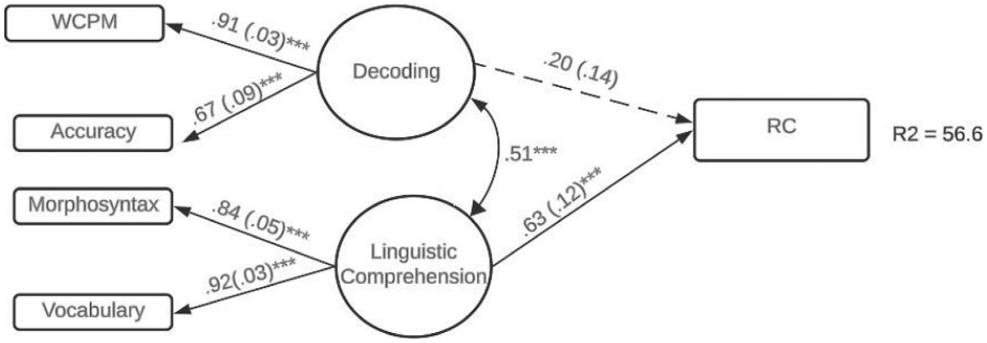

FIGURE 1 | Standardized parameter estimates for the SEM model ($N=42$). Standard errors are in parentheses. ***$p \leq 0.001$. Solid lines indicate significant relationships ($p \leq 0.001$); dashed lines indicate non-significant relationships ($p > 0.05$). RC, reading comprehension; WCPM, words correct per minute.

TABLE 4 | Descriptive statistics for pre-reader ($N=32$).

	n	%	Min	Max	*M* (SD)
Child characteristics					
Gender					
Boys	15	46.9			
Girls	17	53.1			
Spanish school grade					
Pre-kindergarten	15	46.9			
Kindergarten	15	46.9			
First grade	2	6.3			
Generation					
Year 1	27	84.4			
Year 2	5	15.6			
Age	32		4.69	6.97	5.86 (0.56)
Linguistic comprehension					
Receptive vocabulary (max = 125)	32		2	79	41.91 (20.03)
Morphosyntax (max = 49)	32		0	43	18.88 (14)
Decoding precursors					
PA (max = 10)	32		0	10	5.97 (2)
LS (max =25)	32		0	23	13.03 (7.79)
LR (max = 54)	32		0	50	24.41 (16.84)

Abbreviations: LR, letter recognition; LS, letter-sound correspondence; PA, phonemic awareness; WCPM, words correct per minute.

Given this strong correlation, we used a chi-square difference test to compare it to a simpler one-factor model (Kelloway 2017), achieved by constraining the correlation between factors to 1.0. Like the two-factor model, the one-factor model demonstrated an excellent fit to the data (RMSEA = 0, 90% CI [0, 0.2]; SRMR = 0.06; CFI = 1; TLI = 1). The chi-square difference ($\Delta_{\chi^2} = 2.27$, $\Delta_{df} = 1$) was not significant, indicating that the two-factor model, with its additional parameters, does not improve the model fit. This finding suggests that the one-factor model provides a comparable fit to the data as the two-factor model. We also examined parsimonious fit indices, namely the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), which assess the cost–benefit trade-off between fit and complexity (Kelloway 2017). The AIC values for both models were nearly identical (1108.209 for the two-factor model vs. 1108.477 for the one-factor model), while the BIC slightly favored the one-factor model (1128.997 vs. 1130.195). Overall, these comparisons indicate that both models fit the data well, with the simpler model offering comparable explanatory power with one fewer parameter.

TABLE 5 | Bivariate correlations in pre-readers ($N = 32$).

Variable	1	2	3	4	5	6
1. Receptive vocabulary	—					
2. Morphosyntax	0.72***	—				
3. Letter recognition	0.68***	0.56**	—			
4. Phonemic awareness	0.22	0.32	0.27	—		
5. Letter-sound correspondence	0.48**	0.38*	0.64***	0.36**	—	
6. Age	0.36*	0.13	0.41*	0.44*	0.51**	—

*$p < 0.05$.
**$p < 0.01$.
***$p < 0.001$.

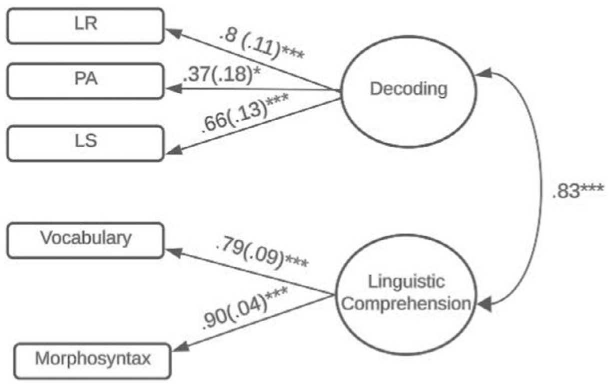

FIGURE 2 | Standardized parameter estimates for the CFA model ($N = 32$). Standard errors are in parentheses; ***$p \leq 0.001$, *$p < 0.05$. LR, letter recognition; LS, letter-sound correspondence; PA, phonemic awareness.

6 | Discussion

We examined cognitive skills related to the development of RC in Spanish HL learners at two stages of reading development (i.e., early readers and pre-readers), employing SVR as an underlying framework. We hypothesized that SVR decoding and linguistic comprehension components uniquely and significantly contribute to RC in Spanish HL early readers. Two distinguishable sets of skills were shown to correspond to linguistic comprehension and decoding. Linguistic comprehension, but not decoding, predicted significant unique variance in RC. However, both components combined explained most of the variance in RC (56.6%). In pre-readers, we found a high correlation between linguistic comprehension and decoding, suggesting that these skills may not be easily separable at earlier stages of reading development.

The finding that only linguistic comprehension predicted unique variance in RC is unexpected considering the predictive power of decoding on RC found across languages (e.g., phonologically transparent and opaque) and decoding measures (e.g., words and nonwords recognition; fluency and accuracy) in prior work (e.g., Baker et al. 2022; Florit and Cain 2011; García and Cain 2014; Hjetland et al. 2020). However, the lack of predictive power for decoding is not an indication of its irrelevance to Spanish RC in our model. The considerable amount of shared variance (26%) between the two SVR components suggests that although not a significant predictor, decoding may influence RC indirectly through its relationship with linguistic comprehension. This implies that successful decoding could support linguistic comprehension processes, which in turn contribute to RC.

It is also important to consider methodological constraints that may have impacted the results. Our small sample size may have constrained our ability to detect smaller effect sizes, such as the 0.20 beta coefficient for decoding predicting RC, potentially impacting the statistical significance of the findings. This highlights the need for larger sample studies to clarify the role of decoding in Spanish HL learners. Nonetheless, the significant effect of linguistic comprehension, even within this limited sample, highlights its substantial and larger impact on RC, compared to decoding.

The strong impact of linguistic comprehension may partly be influenced by the method used for measuring RC. Studies employing open-ended questions (e.g., recalling) show a stronger association between RC and linguistic comprehension. In contrast, cloze or multiple-choice tasks rely more on decoding (e.g., Hjetland et al. 2020; Keenan et al. 2008; Nation and Snowling 2004), which demands specific information and accurate word reading to a greater extent than free recall. Thus, the test with open-ended questions administered in the current study might have privileged the relationship between linguistic comprehension and RC.

In addition to assessment methods, inherent differences in the nature of linguistic comprehension and decoding skills offer another principled explanation for our findings. Kamhi (2007) asserts that decoding constitutes a narrower skill (i.e., involving a narrow set of knowledge, such as recognition of letters, sounds, and words), such that it can only be taught through explicit instruction. However, linguistic comprehension encompasses a broader range of cognitive skills beyond reading (e.g., language processing based on morphosyntax, vocabulary, and listening comprehension), facilitated by exposure to oral language such as storytelling, conversations, and active listening in everyday informal settings, thus reducing its dependence on explicit formal instruction. Given the proportionately reduced explicit instruction in Spanish for participants in our study compared to monolinguals, HL learners rely heavily on linguistic comprehension.

This finding aligns with previous findings from phonologically transparent orthographies, showcasing linguistic comprehension's robust and consistently stronger influence on RC compared to decoding (Cadime et al. 2017; Florit et al. 2022; Tobia and Bonifacci 2015; Torppa et al. 2016). To the extent that Spanish is traditionally categorized as a phonologically transparent orthography, the recent evidence found in Spanish-speaking monolinguals aged 6–10 diverges from our current HL Spanish outcomes. It is important to note that evidence on Spanish RC from cross-linguistic studies remains inconclusive due to the mixed findings yielded by the limited studies available, preventing the establishment of a clear trend.

As we note in the introduction, methodological differences may explain disparities between study findings. Our model measured decoding through accuracy and fluency, aligning with Hoover and Tunmer's (2018) definition of decoding as the ability to rapidly and accurately recognize printed words. In contrast, most reviewed studies on Spanish monolinguals, except for Cubilla Bonnetier and Sánchez-Vincitore (2025), which included separate latent factors for accuracy and fluency, assessed decoding based on different fluency measures (e.g., WCPM, nonword and word reading fluency or a combination of prosody, accuracy and speed). Studies conducted in phonologically transparent orthographies measured aspects of decoding fluency (e.g., words correct per minute, word-nonword reading fluency) and decoding accuracy (word-nonword recognition), except for Torppa et al. (2016), which only includes a fluency measure. Thus, the variability in decoding measures across studies in Spanish monolingual settings—and generally in SVR studies—challenges the comparability of results.

By employing a latent factor analysis, we investigate the combined contributions of these decoding components rather than their individual contributions. Consequently, different methods of assessing the SVR components can reveal variations and commonalities between Spanish HL learners and monolingual children in Spanish-speaking countries, as well as monolinguals from other phonologically transparent orthographies (see summary Table 6 for our findings). Unfortunately, the paucity of investigations and the prevalence of fluency measures in the few studies available in Spanish limit our ability to draw definitive conclusions.

We also explored how decoding precursors and linguistic comprehension skills are associated in Spanish HL pre-readers. The findings align with our expectation that these two components would be highly correlated in the earliest stages of reading development (i.e., younger children). The hypothesized two-factor model proved to be a good fit for the data, and the strong correlation ($r > 0.8$) between the latent variables of decoding and linguistic comprehension is consistent with findings in the literature (e.g., Hjetland et al. 2020; Language and Reading Research Consortium and Chiu 2018; Storch and Whitehurst 2002). Even so, the amount of shared variance involved in any hypothetical path analysis predicting RC from either decoding or linguistic comprehension would be arguably too high (69%). Moreover, the model comparison analysis revealed that the one-factor model fits the data as well as the two-factor model. This raises questions about the practicality and ecological validity of segregating linguistic comprehension and decoding for concurrent analyses at this early stage of reading development. Nonetheless, we do not regard the two-factor model as invalid or lacking in informative value. Supported by excellent fit indices, the two-factor model provides valuable insights into the nuanced relationship between decoding and linguistic comprehension at this early stage. Furthermore, it facilitates meaningful comparisons with early readers' data, enhancing our understanding of developmental trajectories.

The correlation of 0.55 found in our study for early readers falls within the range reported in Spanish studies involving elementary school children, which varies from 0.46 to 0.78 (Camarillo Salazar et al. 2021; Goodrich and Namkung 2019; Wagley et al. 2022). While there is no existing evidence for this association among pre-readers in Spanish, both our study and research in English (Hjetland et al. 2020; Language and Reading Research Consortium and Chiu 2018; Storch and Whitehurst 2002) found a high correlation in this group (preschoolers).

Notably, the correlation between decoding and linguistic comprehension was weaker in our sample of HL readers than in pre-readers. This outcome supports prior longitudinal English research, demonstrating the gradual disassociation between decoding and linguistic comprehension as children become skilled readers (Kendeou, van den Broek, et al. 2009; Language and Reading Research Consortium and Chiu 2018).

In English studies, the decoding-linguistic comprehension relationship in the pre-reading stage has been attributed to the strong links between oral language knowledge, particularly vocabulary, and decoding precursors, such as phonological awareness and letter knowledge (Catts et al. 2015; Foorman et al. 2015; Gorman 2012; Hipfner-Boucher et al. 2014; Lerner and Lonigan 2016). This association has also been observed in Spanish-English bilingual children ages 9–13, where English word reading strongly correlated with vocabulary and morphological awareness (Gottardo et al. 2018).

In the current study, Spanish vocabulary strongly correlated with letter recognition and letter-sound correspondence. Likewise, Spanish morphosyntax demonstrated strong correlations with letter recognition and moderate correlations with letter-sound correspondence. This pattern suggests a shared characteristic in the early stages of reading acquisition in both languages, emphasizing the importance of reinforcing simultaneous instruction across emergent oral language and code-related skills.

In keeping with research literature mandates for a closer examination of what constitutes linguistic comprehension in phonologically transparent orthographies (e.g., Baker et al. 2022; Florit and Cain 2011) and English (e.g., Language and Reading Research Consortium and Chiu 2018), we evaluated the impact of two foundational oral language measures, namely morphosyntax and vocabulary, on a latent construct representing linguistic comprehension. Our data supported the predicted models across HL Spanish readers and pre-readers. Vocabulary, in addition to morphosyntax, exhibited substantial loadings onto a unified factor, contributing significantly to the variance explained by the latent construct of linguistic comprehension. Our findings and outcomes from previous studies employing latent variable modeling, therefore, provide compelling evidence

TABLE 6 | Summary table illustrating the relative contribution of decoding versus linguistic comprehension to RC at different stages of development.

Readers	Language	Early school years (~5–9 years old)	Later school years (~9 years old and older)	Sources
Speakers of phonologically transparent orthographies	Phonologically transparent orthographies (e.g., French, Spanish, Dutch, Greek, Finnish)	LingC > Dec. accuracy LingC > Dec. fluency Except for Florit and Cain's (2011) study where LingC < Dec. fluency	LingC > Dec. accuracy LingC > Dec. fluency	Cadime et al. (2017); Florit and Cain (2011); Florit et al. (2022); Tobia and Bonifacci (2015); Torppa et al. (2016)
Spanish monolinguals in monolingual settings	Spanish	LingC > Dec. fluency Not enough data for decoding accuracy	LingC > Dec. fluency Not enough data for decoding accuracy	Baker et al. (2022); Camarillo Salazar et al. (2021); Kim and Pallante (2012); Sánchez-Vincitore et al. (2022)
English monolinguals	English	LingC < Dec. accuracy/fluency No separate analysis predicting RC were done for accuracy and fluency	LingC > Dec. accuracy/fluency	Florit and Cain (2011); Hjetland et al. (2020); Ripoll Salceda et al. (2014)
Spanish HL learners (results from the present study)	Spanish	LingC > Dec (accuracy and fluency combined)	No data available	

Note: The symbols "<" and ">" denote which component exerts greater or lesser influence on RC.
Abbreviations: Dec., decoding; LingC, linguistic comprehension.

for vocabulary and morphosyntax's role in the complex knowledge involved in linguistic comprehension (e.g., Catts et al. 2015; Gottardo and Mueller 2009; Hjetland et al. 2020; Language and Reading Research Consortium and Chiu 2018).

The promising results of this study are linked to potential limitations. First, the small sample size reduces our analyses' statistical power and can result in larger standard errors, which can pose challenges for the generalizability of the findings. In the context of SEM, it is important to note that goodness-of-fit statistics, which are to a certain degree sensitive to sample size, primarily apply to the measurement model in this study. This is because the structural model used here—including two latent factors predicting RC—is a saturated model, meaning it accounts for all possible relationships among latent variables, and thus, by definition, fits the data perfectly. Nonetheless, the selected indicators for each latent variable are grounded in strong theoretical foundations, which helps ensure robustness in the analysis.

In light of the variability in language competencies among HL speakers within any given HL community (Montrul 2016; Polinsky 2015), the outcomes in our sample may not be representative of Spanish HL in other US regions. Furthermore, this study's sample is unique not only due to its relatively small Southeast Michigan Latino/*x* population (5.6%; DATA USA 2022) but also because of the limited academic and social support for HL Spanish in the United States; it is rare to have access to a critical mass of young HL participants with sufficient Spanish reading proficiency for early literacy tests.

Second, the study is concurrent, not longitudinal. Thus, while it is the first step in examining snapshots of Spanish HL reading stages more methodologically, it does not allow assessing developmental changes in the relationship between the studied components and the outcome measure over time. Third, we followed the standard treatment of orthographic transparency as a binary characteristic, which likely led to inaccurate comparisons owing to the traditional classification that languages are either "all" phonologically transparent or "all" opaque. For instance, numerous "non-phonologically transparent" silent letters and dialectal variations in Spanish pose challenges for early Spanish-language readers and writers. While the question of language transparency categorization is beyond the scope of the current study, we view reworking the conventional binary grouping into a continuum as a worthwhile endeavor.

Moreover, we recognize the need for in-depth studies across different languages to explore how grapheme-phoneme mapping influences the decoding-RC relationship, especially in multilingual contexts like those of Spanish HL learners. Lastly, including a measure of listening comprehension would have provided a more comprehensive understanding of linguistic comprehension and its contribution to RC.

7 | Conclusion and Future Directions

Can the SVR inform the study of RC in young Spanish HL learners? Our findings demonstrate that it can, but only with key considerations. Initially developed and evaluated in English

(Hoover and Gough 1990) with most evidence from English monolingual populations, the SVR is valuable for examining proximal aspects in RC development. However, applying the SVR to unique and diverse populations like HL learners requires attention to population- and language-specific features. The diversity of skills assessed and variations in measurement methods remain ongoing challenges in SVR research, as noted by Storch and Whitehurst (2002). Using methodologies such as latent variable modeling has highlighted the complexity of skills within the SVR components. Ongoing replication efforts are crucial to achieve more precise and consistent conceptualizations and operationalizations of all SVR components.

7.1 | Educational Implications

Our findings indicate that Spanish HL learners will benefit from oral language instruction, particularly targeting Spanish vocabulary and morphosyntax. By fostering HL home cultural language and literacy practices, such as storytelling, shared reading, and interactive conversations, educators can leverage these valuable assets to enhance school instructional practices and significantly benefit these learners. The evidence from this study highlights the critical role of vocabulary and morphosyntax in Spanish RC, providing a foundation for more specialized and targeted instruction that focuses on these specific areas.

However, while emphasizing the importance of oral language development, it is crucial to maintain concurrent and consistent decoding instruction. This recommendation stems from our identification of the mutual influence between decoding and linguistic comprehension precursors among HL pre-readers and the significant shared variance between decoding and linguistic comprehension in explaining RC in readers. Even in our unique sample of Spanish HL learners who, unlike the typical US Spanish HL child, are able to receive support with limited formal instruction in Spanish through a 25-week Saturday program each academic year, ongoing decoding instruction remains vital.

Assessing HL skills early is key to identifying potential challenges with RC in both the HL and English, allowing for timely and targeted interventions. Research shows that strong Spanish HL skills support RC across both languages (Melby-Lervåg and Lervåg 2011; Proctor et al. 2017; Wagley et al. 2022). Detecting difficulties in the SVR components at an early stage is particularly important for recognizing English-language learners who may be at risk of struggling with comprehension before they become proficient in English (Proctor et al. 2017). As a result, our findings have implications not only for bilingual programs and HL schools but also for regular English schools serving Spanish HL students learning English as a second language.

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Disclosure

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Ethics Statement

This study complies with all ethical protocols and has IRB approval—University of Michigan HUM00038234.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Research data are not shared.

Endnotes

[1] Using composite measures, in Step 1, for linguistic comprehension the beta coefficient was $\beta = 0.64$, $p < 0.001$. In Step 2, after including decoding, the coefficient for linguistic comprehension was $\beta = 0.53$, $p < 0.001$, and for decoding, $\beta = 0.28$, $p = 0.045$, explaining 48% of the variance in RC. The similar beta coefficients across methods underscore the robustness of our findings.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.